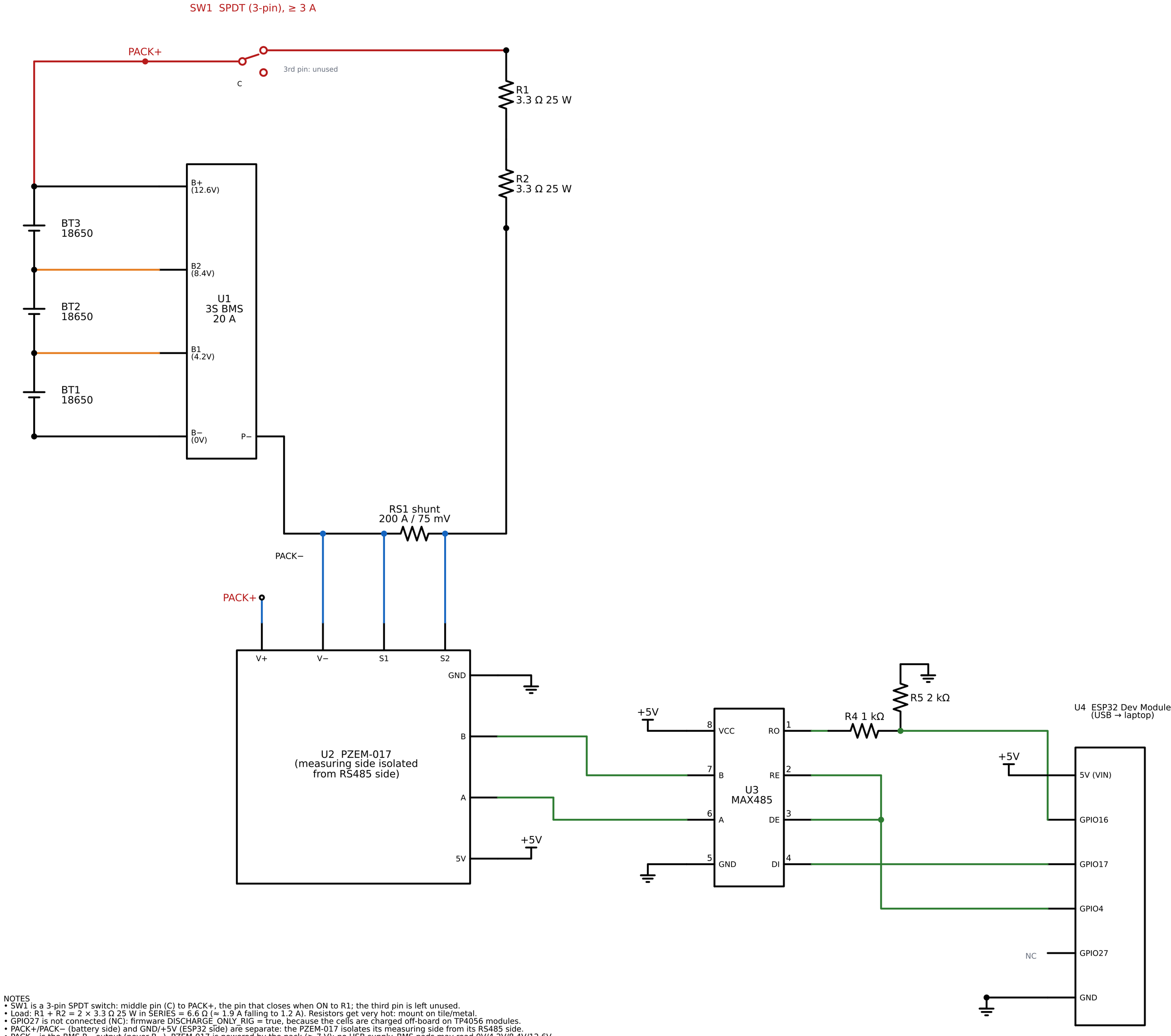

NOTES
• SW1 is a 3-pin SPDT switch: middle pin (C) to PACK+, the pin that closes when ON to R1; the third pin is left unused.
• Load: R1 + R2 = 2 × 3.3 Ω 25 W in SERIES = 6.6 Ω (≈ 1.9 A falling to 1.2 A). Resistors get very hot: mount on tile/metal.
• GPIO27 is not connected (NC): firmware DISCHARGE_ONLY_RIG = true, because the cells are charged off-board on TP4056 modules.
• PACK+/PACK− (battery side) and GND/+5V (ESP32 side) are separate: the PZEM-017 isolates its measuring side from its RS485 side.
• PACK− is the BMS P− output (never B−). PZEM-017 is powered by the pack (≥ 7 V): no USB supply. BMS pads may read 0V/4.2V/8.4V/12.6V.
• Firmware: CELLS_IN_SERIES = 3 (voltage, power and energy are sent per cell). A dot marks a connection; crossings without a dot are not connected.